LA UNI × FEME / LEARNING NOTEBOOKS

Workshop: photovoltaic systems

The electricity pathway, load inventory and maintenance.

100 min · suggested format

Understand an off-grid system, identify energy needs and create an initial energy balance. Learn to make decisions, operate equipment and seek technical support with confidence.

Before starting

Agree voluntary participation, language, timing and what will be recorded. Use the priority chosen by the group. Durations are indicative. Close by returning results and agreeing follow-up.

01 / List what we want to power

20 min. Record each appliance, quantity, rated power and operating hours. Distinguish W (power) from Wh (energy). Example: a 10 W lamp used for 5 hours consumes 50 Wh.

Appliance labels, paper and a calculator.

02 / Follow the energy pathway

20 min. Connect cards for panels, MPPT, batteries, inverter and loads. Explain what is converted and stored. Distinguish daily energy from simultaneous power.

This site's interactive diagram and component cards.

03 / Build a learning energy balance

25 min. Add Wh/day. Use the site's explorer to observe changes in sunlight and demand. Record all assumptions: peak sun hours, losses and consumption. Do not use this exercise as an installation plan.

Load inventory and educational simulator.

04 / Operate and know when to ask for help

20 min. Use de-energised equipment or images to identify indicators, alarms and labels. Agree on usage priorities and technical support. Electrical connections, protection design and work at height require qualified personnel.

Manufacturer manual, logbook and visual inspection protocol.

05 / Explain to make it our own

15 min. In pairs, explain the energy pathway and one care action. Keep an inventory without personal data, questions and review criteria. Schedule a follow-up session.

Suggested evidence: diagram, load balance and a consented recorded explanation.

WORKSHEET

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Load inventory

Daily energy = quantity × W × hours. Example: 2 lamps × 10 W × 5 h = 100 Wh/day. Add measured consumption when available. Startup loads and simultaneous power must be reviewed separately.

Service / appliance	Quantity	W	h/day	Wh/day
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____
_____	_____	_____	_____	_____

A learning balance

Add Wh/day and divide by 1000 for kWh/day. In the simulator: 2.4 kWp × peak sun hours × 0.75. The 0.75 factor is an educational assumption. A positive daily balance does not guarantee night-time service, startup capability or runtime.

Sunlight and loss assumptions: _____
Total consumption and schedules: _____
Unknowns requiring measurement: _____
Questions for qualified personnel: _____

This exercise does not specify connections, protective devices or cable sizing. Do not connect equipment for practice; use cards or de-energised equipment with competent supervision.

Genially / photovoltaic systems (Spanish)

Close and agree

What can we explain in our own words? What would we adapt? What needs confirmation? Keep an authorised record, a responsible role and a next review.